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EX NO: 6	5G NR Frame Structure Analysis with Different Subcarrier Spacing
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Aim

To analyze the 5G NR frame structure by studying the effect of different subcarrier spacings,

slot durations, and OFDM symbols using MATLAB.

Software Required

- MATLAB

Procedure

- Define the supported 5G NR subcarrier spacing values.
- Display the values using MATLAB.
- Plot the subcarrier spacing for different numerologies.

Objective - 1: Analyze Different Subcarrier Spacing Values

MATLAB Program

```
clc;
```

```
clear;
```

```
close all;
```

```
scs = [15 30 60 120 240]; % Subcarrier Spacing (kHz)
```

```
mu = 0:4; % Numerology Index
```

```
bar(scs)
```

```
grid on
```

```
set(gca,'XTick',1:5)
```

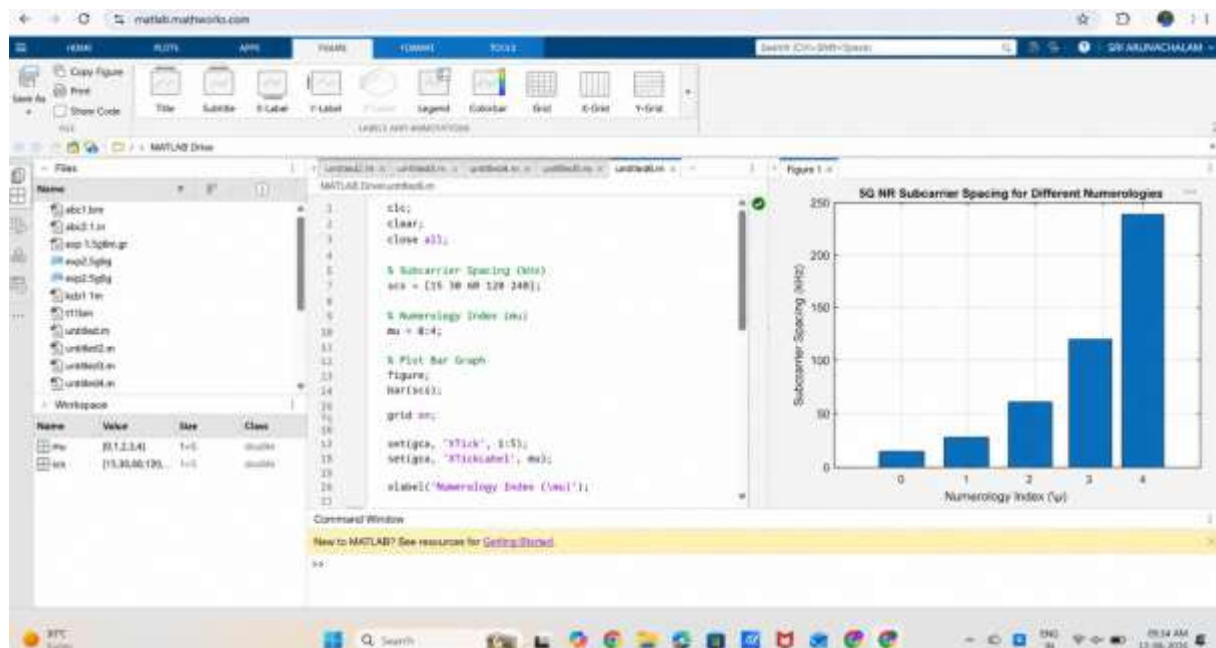
```
set(gca,'XTickLabel',mu)
```

```
xlabel('Numerology Index (\mu)')
```

```
ylabel('Subcarrier Spacing (kHz)')
```

```
title('5G NR Subcarrier Spacing for Different Numerologies')
```

OUTPUT:



Result:

The MATLAB analysis successfully demonstrated the flexible numerology of 5G NR. The results showed that increasing the subcarrier spacing decreases the slot duration while maintaining a constant number of OFDM symbols per slot under the normal cyclic prefix configuration.